

Course Session Selection

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1 Mod selection

1.1 Formal decision problem

Definition 1.1 (COURSE-SESSION-SELECTION). An instance specifies a finite set of courses $\mathcal{C} = \{c_1, \dots, c_N\}$. Course c_i has m_i session types (such as lectures and tutorials). For each type $t \in \{1, \dots, m_i\}$, there are $n_{i,t}$ candidate sessions, represented by intervals

$$I_{i,t,k} = [s_{i,t,k}, e_{i,t,k}], \quad k \in \{1, \dots, n_{i,t}\},$$

where $s_{i,t,k} < e_{i,t,k}$ are rational numbers encoded in binary.

The question is whether there exists a selection function

$$\sigma : \{(i, t) : 1 \leq i \leq N, 1 \leq t \leq m_i\} \longrightarrow \mathbb{Z}_{>0}$$

such that

- (i) $1 \leq \sigma(i, t) \leq n_{i,t}$ for every (i, t) ;
- (ii) all selected intervals are pairwise disjoint: for any $(i, t) \neq (j, u)$,

$$I_{i,t,\sigma(i,t)} \cap I_{j,u,\sigma(j,u)} = \emptyset.$$

Thus exactly one session is selected for every type of every course, and no two selected sessions clash, including sessions of the same course.

Remark 1.2 (Non-overlap and endpoints). The non-overlap requirement is essential: without it, the answer is YES exactly when every session type has at least one candidate. For the closed intervals above, non-overlap means

$$e_{i,t,\sigma(i,t)} < s_{j,u,\sigma(j,u)} \quad \text{or} \quad e_{j,u,\sigma(j,u)} < s_{i,t,\sigma(i,t)}.$$

If back-to-back sessions are allowed, use half-open intervals $[s, e)$ and replace $<$ by $\leq$. The reduction below works under either convention, since intervals intended to be disjoint have positive gaps.

1.2 Membership in NP

A certificate lists one candidate index $\sigma(i, t)$ for each course–type pair. Verify that every index is valid and that every pair of selected intervals is disjoint. With $G = \sum_i m_i$ required types, this uses $O(G^2)$ endpoint comparisons. Each comparison has polynomial bit complexity, so the certificate can be verified in polynomial time. Therefore, COURSE-SESSION-SELECTION $\in$ NP.

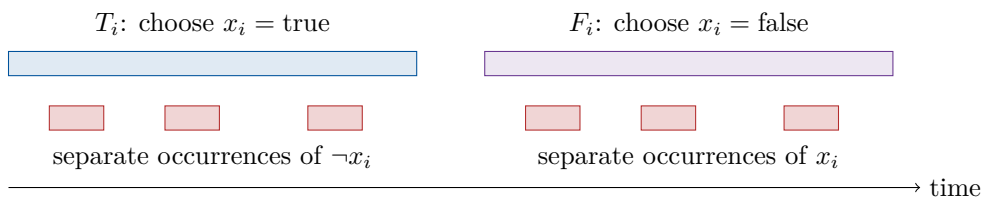
1.3 A direct reduction from SAT

We use SAT in conjunctive normal form (CNF-SAT) as the source problem. The construction also applies directly to 3-SAT. Let

$$\varphi = \bigwedge_{j=1}^p D_j, \quad D_j = (\ell_{j,1} \vee \dots \vee \ell_{j,d_j}),$$

be a formula over variables $x_1, \dots, x_q$, where each literal is x_i or $\neg x_i$. Let $L = \sum_{j=1}^p d_j$ be the total number of literal occurrences. Give every occurrence a distinct index $a \in \{1, \dots, L\}$, even when a literal is repeated. If a clause is empty, the formula is unsatisfiable; map it to a fixed NO instance consisting of two courses, each with one type and the single candidate interval $[0, 1]$. Henceforth, every clause is nonempty.

Idea. Create one course for each variable and one course for each clause, all with exactly one session type. A variable course chooses between two long intervals, T_i and F_i , encoding its truth value. A clause course chooses one short interval for one of its literals. Place that short interval inside the variable interval corresponding to the assignment that makes the literal *false*. Consequently, a clause can choose a literal precisely when that literal is true under the selected variable assignment.



Schematic for one variable. Vertical position only separates the drawings; overlap is determined by their horizontal time intervals.

Variable courses. Set $B = 3L + 3$ and $b_i = 2(i - 1)B$. For each variable x_i , create a course v_i with one session type and exactly two candidates:

$$T_i = [b_i, b_i + B - 1], \quad F_i = [b_i + B, b_i + 2B - 1].$$

Selecting T_i represents setting $x_i = \text{true}$; selecting F_i represents setting $x_i = \text{false}$. All $2q$ variable intervals are pairwise disjoint. Although T_i and F_i do not clash, the requirement to select exactly one session from their common type forces a single truth value for x_i .

Clause courses. For each clause D_j , create a course w_j with one session type and d_j candidates, one for each literal occurrence in D_j . For an occurrence with global index a , define its interval by

$$J_a = \begin{cases} [b_i + B + 3a - 2, b_i + B + 3a - 1], & \text{if the occurrence is } x_i, \\ [b_i + 3a - 2, b_i + 3a - 1], & \text{if the occurrence is } \neg x_i. \end{cases}$$

Since $1 \leq a \leq L$, a positive occurrence lies strictly inside F_i , and a negative occurrence lies strictly inside T_i . Different occurrences in the same variable interval occupy separate slots with positive gaps; occurrences in different variable intervals are also disjoint. Thus *all clause-candidate intervals are pairwise disjoint*, even across different clauses.

The only possible conflicts are therefore

literal represented by J_a	interval conflicting with J_a	assignment making the literal false
x_i	F_i	$x_i = \text{false}$
$\neg x_i$	T_i	$x_i = \text{true}$

The constructed course set is $\mathcal{C}_\varphi = \{v_1, \dots, v_q, w_1, \dots, w_p\}$.

Polynomial size. There are $q + p$ courses, one type per course, and $2q + L$ candidate sessions. All endpoints are nonnegative integers smaller than $2qB = O(q(L + 1))$, so they have $O(\log((q + 1)(L + 1)))$ bits. The construction is computable in polynomial time.

1.4 Correctness and NP-completeness

Lemma 1.3. The formula φ is satisfiable if and only if the constructed COURSE-SESSION-SELECTION instance has a feasible selection.

Proof. ($\Rightarrow$) Suppose φ has a satisfying assignment. For every variable course v_i , select T_i if x_i is true and F_i otherwise. These selected variable intervals are pairwise disjoint.

Every clause has a true literal. For each clause course, select the candidate J_a of one such literal. If the literal is x_i , then J_a lies inside the unselected interval F_i ; if it is $\neg x_i$, then J_a lies inside the unselected interval T_i . In either case, J_a is disjoint from every selected variable interval. All selected clause intervals are also pairwise disjoint by construction. Thus we obtain exactly one session per course type with no conflicts.

($\Leftarrow$) Suppose the constructed instance has a feasible selection. Each variable course selects exactly one of T_i, F_i . Set x_i to true exactly when T_i is selected.

Consider any clause course w_j and its selected interval J_a . If J_a represents x_i , it overlaps F_i , so feasibility forbids selecting F_i . Therefore T_i is selected and x_i is true. If J_a represents $\neg x_i$, the same argument forbids T_i , so F_i is selected and $\neg x_i$ is true. Hence every clause has a true literal, and the extracted assignment satisfies φ . $\square$

Theorem 1.4. COURSE-SESSION-SELECTION is NP-complete, even when every course has exactly one session type, every type has at most three candidate sessions, and all endpoints are nonnegative integers.

Proof. The problem belongs to NP as shown above. The polynomial-time construction and the correctness lemma give the direct reduction

$$\text{CNF-SAT} \leq_p \text{COURSE-SESSION-SELECTION}.$$

This proves NP-hardness. For the stated restriction, apply the same construction to 3-SAT: each variable type has two candidates and each clause type has at most three. Thus the restricted problem, and consequently the general problem, is NP-complete. $\square$